

Primary sources & further reading

Part 15 —  AI Evaluation · 15-ai-evals/papers/PAPERS.md · 15-ai-evals/resources/RESOURCES.md



Papers & Primary Sources — AI Evaluation

Read the three starred papers first — they are the benchmark canon, the judge-bias canon, and the statistics canon respectively. The rest are organized by the section of the notes they back up.

The canon (read in this order)

#	Paper	Year	Why it matters	Link
1	Measuring Massive Multitask Language Understanding (MMLU) — Hendrycks et al.	2020	The benchmark you'll see quoted more than any other. Read it to know exactly what its 57 subjects do and don't cover before quoting its number.	arXiv:2009.03300
2	★ Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena — Zheng et al.	2023	The paper that names and measures position bias, verbosity bias, and self-enhancement bias in LLM judges, and shows GPT-4-as-judge reaches ~80%+ agreement with humans on MT-Bench. The direct source for <code>code/judge_bias_and_calibration.py</code> 's bias catalog.	arXiv:2306.05685
3	★ Holistic Evaluation of Language Models (HELM) — Liang et al.	2022	Multi-metric, multi-scenario evaluation as the honest alternative to a single leaderboard number; the methodology behind "report accuracy, calibration, robustness, fairness, and cost together, not just one."	arXiv:2211.09110
4	★ The Hitchhiker's Guide to Testing Statistical Significance in NLP — Dror, Baumer, Shlomov & Reichart	2018	Surveyed a year of ACL/TACL papers and found most skipped significance testing entirely. The practical protocol for choosing the right test — the paper <code>code/statistical_rigor_for_evals.py</code> 's McNemar/bootstrap choices are built on.	ACL Anthology P18-1128 · companion material: arXiv:1809.01448

Benchmark literacy

Paper	Year	One-line takeaway	Link
Evaluating Large Language Models Trained on Code (Codex / HumanEval) — Chen et al.	2021	Introduces HumanEval and <code>pass@k</code> ; read it to see how narrow the original 164 hand-written problems really are.	arXiv:2107.03374
SWE-bench: Can Language Models Resolve Real-World GitHub Issues? — Jimenez et al.	2023	Real GitHub issues, real repos, tests as the pass/fail oracle — the credible successor to toy code benchmarks, with its own narrowness (12 popular Python repos).	arXiv:2310.06770
GPQA: A Graduate-Level Google-Proof Q&A Benchmark — Rein et al.	2023	448 PhD-level science questions; also reports the striking gap between domain experts (~65%) and skilled non-experts with web access (~34%) — a good citation for "expertise, not search, is the bottleneck."	arXiv:2311.12022

Benchmark contamination

Paper	Year	One-line takeaway	Link
Time Travel in LLMs: Tracing Data Contamination in Large Language Models — Golchin & Surdeanu	2023	The "guided instruction" technique: prompt the model with a dataset name plus a truncated instance and see if it completes the rest verbatim. Found GPT-4 contaminated on several standard datasets.	arXiv:2308.08493
Pretraining on the Test Set Is All You Need — Schaeffer	2023	A deliberately absurd 1M-parameter model trained directly on benchmark data "beats" every real foundation model on the leaderboards it was trained on. The cleanest possible demonstration of why a leaderboard number alone proves nothing about capability.	arXiv:2309.08632

LLM-as-judge and human evaluation

Paper	Year	One-line takeaway	Link
Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena — Zheng et al.	2023	(See canon above — the bias catalog paper.)	arXiv:2306.05685
Holistic Evaluation of Language Models (HELM) — Liang et al.	2022	(See canon above.)	arXiv:2211.09110

Statistical rigor

Paper	Year	One-line takeaway	Link
The Hitchhiker's Guide to Testing Statistical Significance in NLP — Dror et al.	2018	(See canon above.)	ACL Anthology P18-1128
Show Your Work: Improved Reporting of Experimental Results — Dodge et al.	2019	Variance across random seeds is often larger than the "improvement" a paper reports; argues for reporting expected validation performance as a function of compute budget, not a cherry-picked best run. Directly relevant to the p-hacking section of <code>code/statistical_rigor_for_evals.py</code> .	arXiv:1909.03004

How to read an eval paper (30 minutes, 3 passes)

1. **Pass 1 (5 min)**: what exactly is being measured — read the task definition and one full example, not just the headline metric name.
2. **Pass 2 (15 min)**: how was the reference/gold label produced (human annotation? another model? scraped?), and what's the inter-annotator or inter-judge agreement, if reported at all.
3. **Pass 3 (10 min)**: the limitations section, and specifically whether the authors discuss contamination risk or benchmark saturation. If a benchmark paper has neither, treat every score derived from it a little more skeptically.

Keep a one-paragraph note per paper: what it measures, what it doesn't, and the one number you'd actually want next to any score quoted from it (an interval, an agreement statistic, a contamination check).

Resources — AI Evaluation

Courses & long-form guides (free)

Resource	Why	Link
Hamel Husain — "Your AI Product Needs Evals"	The single best practitioner essay on this whole topic; real case study, no hand-waving	https://hamel.dev/blog/posts/evals/
Hamel Husain — AI Evals notes/course hub	Ongoing collection of eval-focused writing and the paid "AI Evals for Engineers & PMs" course	https://hamel.dev/notes/llm/evals/
DeepLearning.AI — Evaluating and Debugging Generative AI	Short, free, hands-on course from the W&B team on tracking experiments and evaluating LLM apps	https://www.deeplearning.ai/short-courses/evaluating-debugging-generative-ai/
DeepLearning.AI — Building and Evaluating Advanced RAG	RAG-specific evaluation (faithfulness, relevance) with TruLens and LlamaIndex	https://www.deeplearning.ai/short-courses/building-evaluating-advanced-rag/
Stanford HELM project site	The live, running implementation of the HELM paper's holistic multi-metric evaluation	https://crfm.stanford.edu/helm/
LMarena (formerly Chatbot Arena) leaderboard + methodology	Read the methodology page, not just the leaderboard, before citing an Arena rank	https://lmarena.ai/ (rebranded to https://arena.ai/ in 2026)

Eval frameworks & repositories

Repo	What's inside
https://github.com/openai/evals	OpenAI's open eval framework and a large registry of community-contributed eval templates
https://github.com/confident-ai/deepeval	Pytest-style LLM eval framework: G-Eval, hallucination, answer relevancy, RAGAS-compatible metrics, CI-friendly
https://github.com/explodinggradients/ragas	The reference RAG evaluation library — faithfulness, context precision/recall, answer relevance; owned in depth by <code>./08-rag/</code>
https://github.com/promptfoo/promptfoo	CLI/CI-first prompt and RAG testing, matrix comparisons across models, and built-in red-teaming/vulnerability scanning
https://github.com/EleutherAI/lm-evaluation-harness	The standard open harness for running academic benchmarks (MMLU, GPQA, etc.) against any model — pairs with the benchmark-literacy section of note 01
https://github.com/stanford-crfm/helm	The HELM paper's actual implementation
https://github.com/langchain-ai/langsmith-sdk	The SDK behind <code>evaluate()/aevaluate()</code> covered in <code>./16-ai-tech-stack/tracks/langsmith/</code>
https://github.com/langfuse/langfuse	Open-source, self-hostable alternative with a thinner but growing eval layer

Cheat sheets & reference docs

- OpenAI Evals design doc and registry format: <https://github.com/openai/evals/blob/main/docs/build-eval.md>
- Cohen's kappa, worked through carefully (general reference, not LLM-specific): https://en.wikipedia.org/wiki/Cohen%27s_kappa
- Wilson score interval derivation: https://en.wikipedia.org/wiki/Binomial_proportion_confidence_interval#Wilson_score_interval
- Also see ../../_shared/cheatsheets/

Communities

- Hamel Husain's "AI Evals" Discord/newsletter community (linked from <https://hamel.dev/>) — the most active practitioner-level discussion of eval craft specifically, as opposed to general LLM-ops
- r/LocalLLaMA and the EleutherAI Discord (shared with Module 04 [04-llm]) — useful for benchmark-methodology discussion whenever a new leaderboard controversy breaks
- LangChain / LangSmith community Discord — practical `evaluate()` and dataset-design questions, tooling side of this module